

A2: TEC Heating and Cooling Slope Analysis

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GitHub: github.com/joshuabub-hub/PHYS39A_repo

Part 4: Combined graph and fits

The plot generated by `Python/plot_steady_state.py` combines the heating and cooling steady-state measurements on one graph. The horizontal axis is signed PWM command (counts; positive for heating, negative for cooling), and the vertical axis is steady-state object temperature (°C). The fits use the full tested ranges: heating PWM **0–45** and cooling signed PWM **–115 to 0**. The PWM 0 measurement is the shared unpowered baseline; it is shown once on the graph and used as the zero-duty endpoint for both linear fits.

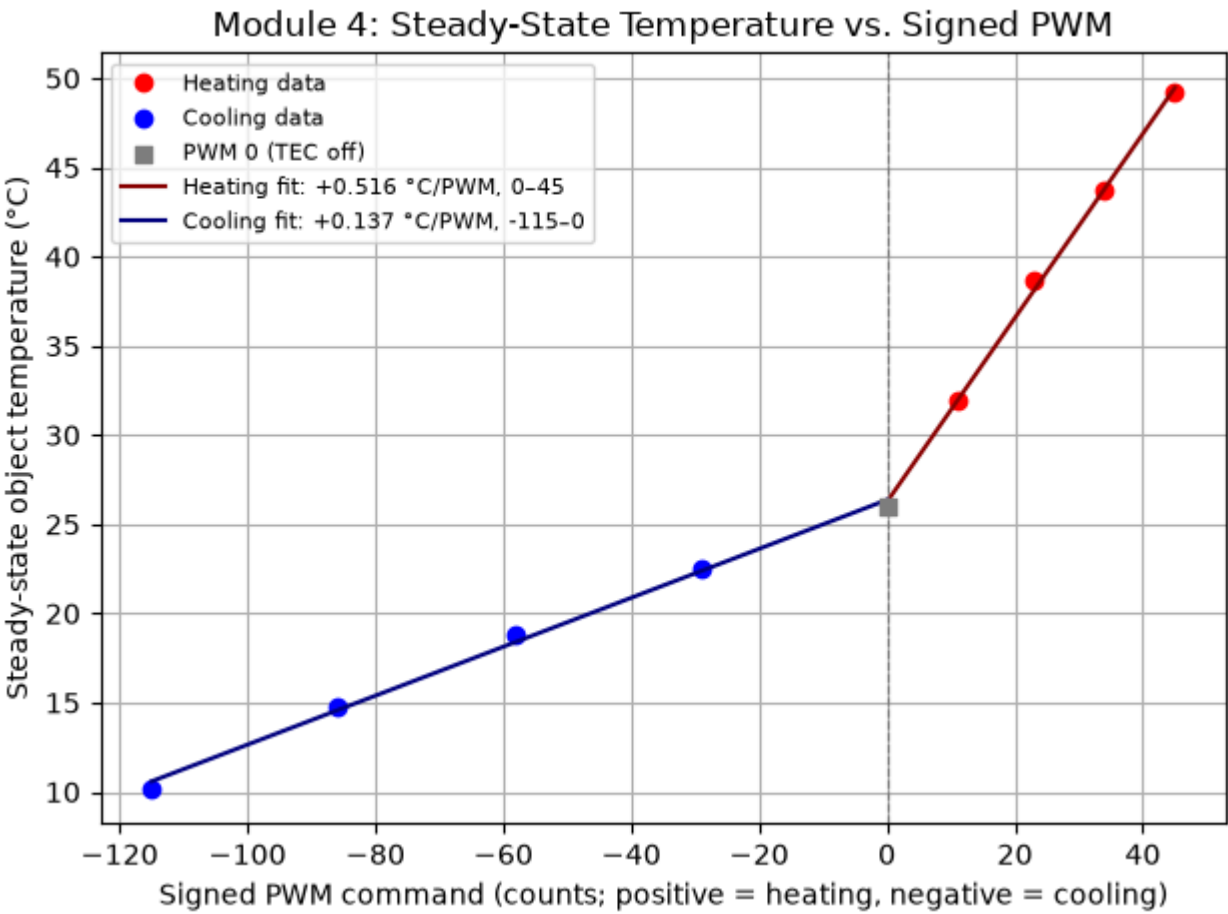


Figure 1. Steady-state object temperature (°C) vs. signed PWM command (counts). Red: heating data and fit; blue: cooling data and fit; gray square: PWM 0 (TEC off) shared baseline.

Branch	Fitting range (PWM counts)	Fitted slope	Slope units
Heating	0 to +45	+0.516	°C per PWM count
Cooling	–115 to 0	+0.137 (magnitude 0.137)	°C per PWM count

Slope ratio: $r = m_h/m_c = 0.516/0.137 \approx 3.76$ (dimensionless).

Run the script from the Module 4 folder with the project dependencies installed. It reads `data/pwm_steady_state.csv`, prints the fitted slopes and their ratio, and saves `data/pwm_steady_state_plot.png`.

Part 5.1: Measured slopes

Ordinary least-squares fits of steady-state temperature against signed PWM over the full tested ranges give:

Quantity	Measured value
Heating slope, m_h	$+0.516\text{ }^\circ\text{C/PWM count}$
Cooling slope magnitude, $m_c = dT/dd_{\text{PWM}} $	$0.137\text{ }^\circ\text{C/PWM count}$
Ratio, $r = m_h/m_c$	3.76

The measured cooling branch is plotted against signed PWM, so its fitted line slope is positive: increasing signed PWM from -115 toward -29 raises the temperature. The table reports its magnitude as requested.

The heating points show no clear systematic curvature. Cooling shows a small bend (the interior points sit slightly above a single straight-line fit), but the residuals remain small relative to the total temperature span (maximum absolute residual about $0.55\text{ }^\circ\text{C}$ for heating and $0.42\text{ }^\circ\text{C}$ for cooling). We use the full measured ranges rather than selecting a narrower interval, so the reported slopes represent average sensitivities over the useful ranges. The 0% row is the same power-off segment for both directions and serves as the shared zero-duty reference; subtle curvature cannot be ruled out from only five points per branch.

Part 5.2: PWM and the slope-ratio model

For a PWM period τ , let current of magnitude I flow during the on-time $D\tau$ and be zero during the remaining $(1 - D)\tau$. Then

$$\langle I \rangle = \frac{1}{\tau} \int_0^\tau I(t) dt = \frac{I(D\tau)}{\tau} = DI, \quad \langle I^2 \rangle = \frac{1}{\tau} \int_0^\tau I(t)^2 dt = \frac{I^2(D\tau)}{\tau} = DI^2.$$

Thus the average Joule heating power, $\langle I^2 \rangle R = DI^2 R$, and the average Peltier heat rate, proportional to $\langle I \rangle = DI$, both scale linearly with D . This assumes an ideal on/off current of fixed magnitude I and a PWM period τ much shorter than the thermal time constant, so the object responds only to the time-averaged heat rates.

Define signed duty cycle $d = u/255$ and $D = |d|$. Let $\dot{Q}_P > 0$ be the full-on Peltier heat rate into the object and $\dot{Q}_J > 0$ the full-on Joule heat rate deposited at the object face. Use

$$\dot{Q}_{\text{TEC}} = d\dot{Q}_P + |d|\dot{Q}_J, \quad C \frac{dT}{dt} = \dot{Q}_{\text{TEC}} - G(T - T_0).$$

Here C is the heat capacity of the object, G the passive thermal conductance to the surroundings, and T_0 the ambient (room) temperature. At steady state, $dT/dt = 0$ and $T = T_0 + \dot{Q}_{\text{TEC}}/G$. For heating, $d \geq 0$ and $|d| = d$, so $\dot{Q}_{\text{TEC}} = d(\dot{Q}_P + \dot{Q}_J)$. For cooling, $d < 0$ and $|d| = -d$, so $\dot{Q}_{\text{TEC}} = d(\dot{Q}_P - \dot{Q}_J)$. Differentiating the steady-state temperatures with respect to signed duty cycle gives

$$m_h = \frac{dT_h}{dd} = \frac{\dot{Q}_P + \dot{Q}_J}{G}, \quad m_c = \frac{dT_c}{dd} = \frac{\dot{Q}_P - \dot{Q}_J}{G}.$$

When $\dot{Q}_P > \dot{Q}_J$, both branch derivatives are positive and $m_h > m_c$. The slopes per signed PWM count each include the same factor $1/255$, which cancels in the ratio $r = m_h/m_c$. Therefore,

$$r = \frac{\dot{Q}_P + \dot{Q}_J}{\dot{Q}_P - \dot{Q}_J} \implies \frac{\dot{Q}_J}{\dot{Q}_P} = \frac{r - 1}{r + 1}.$$

With the full-range measured value $r = 3.76$,

$$\frac{\dot{Q}_J}{\dot{Q}_P} = \frac{3.76 - 1}{3.76 + 1} \approx 0.580.$$

The measured slope ratio therefore implies an object-face full-on Joule heat rate about **0.58 times** the full-on Peltier heat rate, under this model. This requires $r > 1$ (i.e. $\dot{Q}_P > \dot{Q}_J$), which the data satisfy. As a check, if $r = 2$, the formula gives $\dot{Q}_J/\dot{Q}_P = 1/3$.

Part 5.3: Laird data-sheet calculation

The specified module is the Laird Thermal Systems **CP14-127-045-L2-W4.5** (manufacturer part number **58910-501**). Values below are from the "SPECIFICATIONS" table on page 3 of the [Laird datasheet](#) (Revision 00, 06-01-2022), using its $T_{\text{hot}} = 27^\circ\text{C}$ column. That column is the closest match to our rig, whose hot side sits near room temperature (the unpowered object settles at about 26°C).

Datasheet parameter	Value	Meaning and stated condition
R_M (module resistance)	$1.50\ \Omega$	Electrical module resistance; datasheet column $T_{\text{hot}} = 27^\circ\text{C}$.
I_{max}	$8.6\ \text{A}$	Listed as current at ΔT_{max} ; $T_{\text{hot}} = 27^\circ\text{C}$.
$Q_{c,\text{max}}$	$71.3\ \text{W}$	Maximum heat pumped at the cold side, at $\Delta T = 0$; $T_{\text{hot}} = 27^\circ\text{C}$.
ΔT_{max}	70.5°C	Maximum hot-to-cold temperature difference at $Q_c = 0$; $T_{\text{hot}} = 27^\circ\text{C}$.

For the usual approximately equal split of Joule heat between the module's two faces, the object's-face Joule rate at the listed current is

$$\dot{Q}_{J,\text{max}} = \frac{1}{2} I_{\text{max}}^2 R_M = \frac{1}{2} (8.6\ \text{A})^2 (1.50\ \Omega) = 55.47\ \text{A}^2\ \Omega = 55.47\ \text{W}.$$

At $\Delta T = 0$, the cold-face heat pumped is the Peltier contribution minus the cold-face share of Joule heating. Thus, using the datasheet maximum-current values in the requested lumped model,

$$\dot{Q}_{P,\text{max}} \approx Q_{c,\text{max}} + \dot{Q}_{J,\text{max}} = 71.3\ \text{W} + 55.47\ \text{W} = 126.77\ \text{W}.$$

This gives the datasheet-based maximum-current slope-ratio prediction

$$r_{\text{Laird,max}} = \frac{\dot{Q}_{P,\text{max}} + \dot{Q}_{J,\text{max}}}{\dot{Q}_{P,\text{max}} - \dot{Q}_{J,\text{max}}} = \frac{126.77\ \text{W} + 55.47\ \text{W}}{126.77\ \text{W} - 55.47\ \text{W}} = \frac{182.24}{71.30} \approx 2.56 \quad (\text{dimensionless}).$$

The corresponding Joule-to-Peltier ratio is $\dot{Q}_{J,\max}/\dot{Q}_{P,\max} = 55.47/126.77 \approx 0.44$.

The calculation is an estimate using datasheet headline values: the table defines $I_{\max}$ at $\Delta T_{\max}$, while $Q_{c,\max}$ is specified at $\Delta T = 0$. These extrema are not explicitly guaranteed to be a single simultaneous operating point, so the resulting ratio should not be treated as a precision prediction.

Part 5.4: Compare the ratios

The measured full-range ratio is $r \approx 3.76$, versus the datasheet-based estimate $r_{\text{Laird},\max} \approx 2.56$. The measured value is about 47% higher; equivalently, the measured $\dot{Q}_J/\dot{Q}_P \approx 0.58$ exceeds the datasheet-based 0.44. Because r rises steeply as $\dot{Q}_J/\dot{Q}_P$ approaches 1, a modest difference in that ratio produces a large difference in r . Exact agreement is not expected: the datasheet estimate combines its rated maximum-current and maximum-capacity values, whereas the experiment uses a different heatsink, contact, ambient temperature, and operating range. Module resistance and thermoelectric coefficients also vary with temperature, and the H-bridge/supply may change the current from the ideal model.

Operating current alone does not explain the direction of the difference: Joule heating scales as I^2 while the Peltier term scales as I , so a rig running below the datasheet's $I_{\max}$ would have a smaller $\dot{Q}_J/\dot{Q}_P$ and a lower r , not a higher one. Other effects, such as heat leakage, driver voltage drop, or the datasheet's $Q_{c,\max}$ and $I_{\max}$ not being a single operating point, must contribute.

In particular, a 100% PWM command ($D = 1$) only means the controller holds the output on continuously. It does **not** regulate the TEC to $I_{\max}$: actual current is set by supply voltage, module resistance, driver voltage drop, wiring, and thermal operating conditions. Conversely, the Laird $I_{\max}$ is a datasheet rating defined at its specified test conditions, not a current guaranteed by this rig at PWM 255.

Passive heat conduction (also Part 5.4)

The passive heat-flow term into the object is $-G(T - T_0)$. If the object is hotter than the room ($T > T_0$), passive heat flows from the object to the room. If the object is colder ($T < T_0$), heat flows from the room into the object. This approximately symmetric conduction opposes either deviation from ambient: it removes heat in the heating branch and adds heat in the cooling branch. The unpowered steady state ($\approx 26^\circ\text{C}$ at PWM 0) is the reference temperature T_0 at which this term vanishes. In the linear model it contributes the same conductance G to both branch slopes, so it cannot by itself explain unequal slope magnitudes; the difference comes from the branch-dependent Peltier and Joule terms.